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Hybrid Vision Transformer Framework for Brain Tumor Detection and Classification

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***Abstract*— Brain tumor, a common type of uncontrolled cell growth in the central nervous system, is very difficult to medically diagnose and treat. Early diagnosis is critical for effective treatment. This paper aims to improve the detection and classification of brain tumor in the Magnetic Resonance Imaging (MRI) scan using a vision transformer (ViT) based innovative framework. This paper created a generic image processing pipeline with resize, augmentation, and normalization, fine-tuning four ImageNet-pretrained backbones (ViT, DeiT, Swin, and CaiT) by freezing most of the layers, while only unfreezing the last few blocks and adding a new classification head. AdamW and cosine learning-rate scheduler are used and mixed-precision calculations. The predictions are combined with a soft voting ensemble, which takes the average of the model logits. Performance is presented in terms of accuracy, precision, recall, and F1-score. ViT can achieve an accuracy of 97.25%; the all-models ensemble achieves an accuracy of 95.88%. The best performing configuration is a two-model ensemble (ViT+DeiT), which resulted in 97.71% accuracy, 97.70% precision, 97.57% recall, and a 97.59% F1-score, outperforming the single backbones while being lightweight. The framework is evaluated and the experimental results show that the proposed framework can achieve robust yet interpretable MRI brain tumor classification and that the trained transformer encoders and selective ensembling are well-suited for clinical triage & deployment.**

Keywords*— Brain tumor classification, deep learning, vision transformers, ensemble learning.

I. INTRODUCTION

The human brain is composed of billions of neurons and their synapses which collectively control vital functions in the human body. Like other organs, it is prone to disease; one is the formation of brain tumors. These tumors result from the abnormal proliferation of cells in various regions of the brain, causing serious neurological complications and significantly reducing the quality of life of patients [1]. Brain tumors have long been known to be one of the most fatal disease entities in the world, having a significant impact upon mortality in all age groups. They are also very difficult to diagnose and treat [2]. Brain tumors can present in a number of forms, including glioma, meningioma, or pituitary tumors, or no tumor at all. With a total of nearly 120 types of entorhinal cortices found to date, its functional heterogeneity and size range, as well as the intricate anatomy of the brain, means that unambiguous detection is a challenging task [3]. Automated brain tumor detection and classification are crucial to provide an early diagnosis and achieve the best patient outcome, as manual review of big medical image datasets is usually time consuming and expensive [4]. Despite the importance of physician expertise, manual segmentation of 3D MRI scans is still a difficult task, especially for brain tumor delineation in multimodal MRI. Automated classification techniques allow to overcome the user variability as the results are objective and reliable [5]. Over the years, different machine learning techniques and imaging modalities have been adopted in brain tumor detection [6,7]. Recently, a new branch of machine learning, known as deep learning, has emerged as a potential candidate for brain tumor classification and segmentation to provide accurate and reliable assistance for computer-aided diagnosis to determine the presence of tumor, its location, and its type.

Despite these advances, however, current models face difficulties in terms of efficiency, robustness, and interpretability, which restricts their clinical applicability. In order to fill these gaps, this study explores the use of a Vision Transformer (ViT) based framework to improve brain tumor detection and classification with better accuracy and reliability.

- Our study is based on a curated dataset of multi-source MRI data standardized to four disease classes to facilitate reliable brain tumor analysis and diagnosis.
- Developed a hybrid ViT–DeiT ensemble that achieved the top results on our data, improving detection and classification.
- In addition, the study used targeted pre-processing and fine-tuning with AdamW using cosine LR and mixed precision for performance optimization.
- DL baselines are superseded by an ensemble for brain tumor detection in terms of accuracy, precision, recall and F1.

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II. Related Work

In this section, we discussing the present state of the art of brain tumor diagnosis based on DL detection and classification and define the problem statement accordingly.

To derive rich representations from brain MRI scans the work in [8] tested several off-the-shelf deep networks InceptionResNetV2, NASNet-Large, DenseNet121 and Xception. The experiments were based on the Brain Tumor Detection 2020 (BR35H) dataset that contains 1500 images with tumor-free brain and 1500 images with tumor in brain. Among these models, InceptionResNetV2 showed the best results, with a precision of 99.68%.

H. A. Shah et al. [9] fine tuned an EfficientNet based architecture and tested on a dataset of 3762 brain MRI images. Model performance was evaluated on standard metrics that count correct predictions vs. incorrect predictions. Our system successfully classified 338 images as non-tumorous, where five images were incorrectly classified from this class.

In [10], the authors proposed a three-stage preprocessing pipeline of suppressing distracting non-brain structures, suppressing noise in MR images and applying histogram equalization and an effective deep learning diagnostic framework. Their proposed DCNN achieved class wise AUCs in between 94% and 96% on a data set consisting of 3394 images.

M. Rizwan et al [11], a Gaussian Convolutional Neural Network (GCNN) was proposed for classification of multiple brain tumor types using two brain tumor datasets. The method obtained 99.8% accuracy on Tianjin Medical University dataset and 97.14% accuracy on the TCIA dataset. The GCNN has four convolutional layers, and its learning behavior is stable and there is no obvious overfitting or underfitting.

According to Ghosal et al. [12], for the multi-class classification of brain tumors, a ResNet-101 Squeeze-and-Excitation CNN model was proposed. The pipeline first extracts the tumor ROI from T1-weighted contrast enhanced slices of MR, then zero-centering, intensity normalization and data augmentation are performed before training the SE-ResNet-101 using Adam. Experiments on the well known Figshare dataset of 3064 images of 233 patients of glioma, meningioma and pituitary tumor report overall accuracy 93.83 with class-wise accuracies of 98.67 for glioma and 91.81 and 91.03 for meningioma and pituitary, outperforming previous SVM and DNN baselines on the same data.

In [13], convolutional dictionary learning with local constraints (CDLLC) was investigated, where a CNN backbone is used that terminates on an AlexNet style head and a softmax classifier. Local structure was enforced by creating KNN graphs and a sensitivity analysis of important CDLLC parameters was performed. Public datasets: Cheng (3064 images, three classes) and REMBRANDT (130 images of patients with astrocytoma, oligodendroglioma, glioblastoma multiforme and other types of tumor). Experiments were performed on the complete Cheng set and 1000 images from REMBRANDT, and performance is shown as the average of 10 runs.

The previous study and its limitations are summarized in table 1. To overcome these gaps, in this work, deep learning models are evaluated and stacking-based combinations are used to choose the best suited predictor. The approach can be extended to large MRI data sets, is more robust through comparative validation, and can facilitate brain tumor detection across several clinically relevant categories.

Table 1: Related publications along with boundaries.

References	Models	Data set Size	Limitations
S. Asif et al. 2022 [8]	Xception, NasNet Large, DenseNet121	3000	Summary of Hyperparameters table seems unnecessary as all the variables are kept same for all the models.
H. A. Shah et al.2022 [9]	VGG16, InceptionV3, Xception, ResNet50	3762	Accuracy and loss curves for the compared models could also be given in the paper
X. Gu et al. 2021 [10]	VGG16, VGG19, CNN-SVM, DCNN	3304	AUCs of other compared models could have been shown along with the proposed model.
M. Rizwan et al 2022 [11]	Gaussian Convolutional Neural Network (GCNN),	3064, 516	Comparison is not accurate
Ghosal et al. 2022 [12]	SE-ResNet-101	3064	Evaluated only on a single 2D CE-MRI dataset without external/clinical validation, so the model's real-world generalization is uncertain.
A. S. Musallam 2022 [13]	CDLLC	4064	The authors did not implement any existing models to compare results with the proposed model

III. Materials And Methods

Figure 1 is the flowchart for proposed optimized method for brain tumor classification with ViT. The description of the methodological procedures adopted to use classifiers and related techniques applied in this study is provided. The figure outlines each step of the pipeline, from data preprocessing to training the model and assessing the performance.

A. Dataset Collection

The work is based on the dataset provided by Kaggle [14] curator nickparvar that is a combination of brain MRI images from Figshare, SARTAJ dataset, and Br35H. It consists of 7,023 images classified into four classes including glioma, meningioma, pituitary and no tumor. We utilized the original folders which had 5,712 training and 1,311 testing images; with 1,321 train and 300 test images per class (glioma), 1,339 train and 306 test images per class (meningioma), 1,457 train and 300 test images per class (pituitary), and 1,595 train and 405 test images per class (no tumor). Because source images come in several different sizes, we trimmed nonbrain margins and resized all images to a fixed input size prior to model training. Figure 2 shows

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representative MRI slices from each diagnostic group used in this study: no tumor (top left), glioma (top right), meningioma (bottom left) and pituitary (bottom right). Examples are axial and sagittal slices, which correspond to the heterogeneous acquisition conditions of the Kaggle data set. The optical lesions are heterogeneous in spot position, dimensions, and intensity distribution, showing the challenge in the classification task and encouraging a robust preprocessing and modeling. less susceptible to overfitting and more generalizable to unseen data.

B. Data Preprocessing

Data preprocessing is one of the essential tools to make sure that raw brain MRI images are converted into a format acceptable for training deep learning models, especially Vision Transformer (ViT)-based architectures. Since model performance is directly affected by input data quality and consistency, several preprocessing steps are used. These steps have the dual purpose of standardizing the dataset, but also make the dataset more robust,

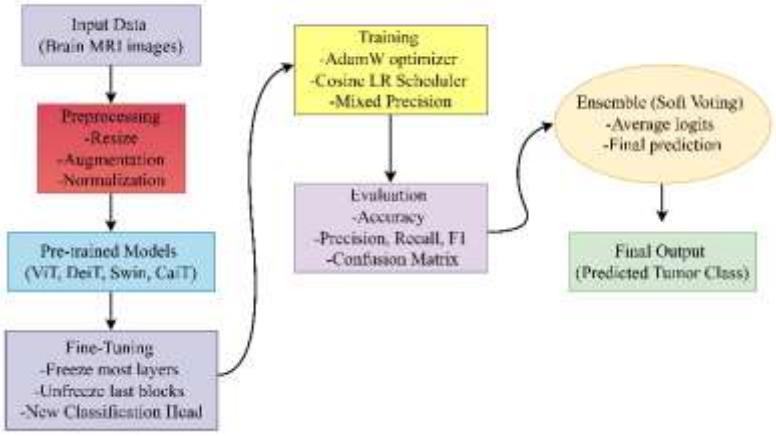


Fig. 1: Workflow of our Proposed Model

1) Dataset Organization

The dataset consists of Training and Testing directories with further subdirectories for each tumor class including, glioma, meningioma, pituitary, and no tumor. This class-based hierarchical directory structure can be used with PyTorch's dataset loaders and auto-assign them labels based on directory names.

2) Grayscale Conversion with 3 Channels

MRI images are naturally grayscale, i.e., they have only intensity channels instead of RGB color channels. However, there are pre-trained transformer architectures (e.g. ViT, DeiT, Swin, CaiT) which expect 3-channel input tensors, because they are trained on ImageNet which contains RGB images. In order to be compatible with the monitor, each single-channel grayscale image is tiled across three channels, thus forming a pseudo-RGB image. This step does not add any new information, but matches the input format of the expected pre-trained model input, so it can be used for transfer learning.

3) Image Resizing

All images are downsampled to a fixed spatial resolution of IMGSIZE X IMGSIZE (e.g., 224 X 224). Vision Transformers break input images into fixed-size patches (e.g., 16 x 16) before these go into the encoder. Therefore, it is necessary to have the input dimension constant for correct patch extraction and positional encoding. The resizing guarantees homogeneity among the dataset, cuts down on CPU overhead and allows fast GPU batch processing.

4) Normalization

In order for the pixel intensity distribution of the MRI images to match the distribution of the ImageNet trained models, normalization with ImageNet's mean and standard deviation values is used. The equation of Normalized Image is represented in equation (1),

$$\text{Normalized Image} = \frac{(I - \mu)}{\sigma} \quad (1)$$

Where, $\mu = [0.485, 0.456, 0.406]$ $\sigma = [0.229, 0.224, 0.225]$.
Normalization ensures:

- The input distribution matches the pre-training distribution, stabilizing training.
- Faster convergence due to standardized input scales.
- Reduced risk of gradient explosion or vanishing

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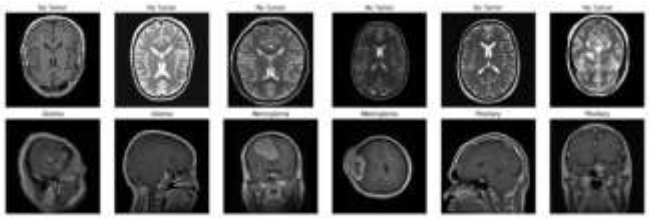


Fig. 2. Representative Brain MRI Samples Across Four Classes.

5) Data Augmentation (Training Only)

To better generalization and to avoid overfitting problem for a limited medical data, we perform stochastic data augmentation on the training split only. The scheme incorporates random horizontal flipping with probability 0.5 to simulate differences in patient orientation, random rotations (up to +-30deg) to mimic the normal scanners, affine perturbations (including translation and shear) to approximate differences in acquisition and subtly shift anatomy, color jitter to vary brightness, contrast, saturation, and hue (implemented on replicated channels so that the model is robust to the presence contrast differences between scans,

even though MRI is grayscale), and random erasing that occludes a rectangular region and forces the network to learn to rely on contextual information and not specific pixels. Collectively, these transformations lead to a variety of versions of the training images, exposing the model to a wide range of plausible scenarios, and reinforcing robustness in clinical settings.

6) Dataset Splitting

The training data set is further divided into training and validation using randomsplit. This ensures that the model is trained using one subset of the dataset while the validation occurs on another unseen subset of data, which gives an unbiased estimate of the generalization output of the model during training.

7) Data Loading

In order to get the most out of the GPU, we set up PyTorch DataLoaders so that the training DataLoader uses data augmentation and shuffles samples to decouple sequential correlations, but the validation and test DataLoaders use only resizing and normalization to make sure evaluation is consistent. The NUMWORKERS parameter determines how many subprocesses will be used for data loading in parallel to minimize I/O bottlenecks, and BATCH_SIZE dictates how many images are processed per iteration balance between memory efficiency and gradient stability.

C. Transformer-based Models

1) DeiT: DeiT (Data-efficient Image Transformer):

DeiT is a data and compute efficient version of the Vision Transformer. It segments an image into patches and passes them to a transformer encoder which captures long-range dependencies across the entire image. By using knowledge distillation between DeiT and an auxiliary CNN teacher (using a special distillation token), DeiT is able to achieve high training translation speed with much less training data and lower computational cost compared to standard ViT implementations. In addition to this core recipe, DeiT is built upon a robust augmentation and optimization pipeline that stabilizes learning on smaller datasets, further reducing the difference between pure transformers and CNNs. The result is a lightweight and competitive classifier that maintains the global-context advantages of self-attention while remaining practical for low-data or resource-limited settings [15].

2) Swin :

The Swin Transformer is a scalable yet strong-performing vision transformer architecture. It constructs a hierarchical (pyramidal) representation by internally computing self-attention in non-overlapping windows and then shifting the windows between layers while preserving the spatial flow of information. The proposed approach captures local details and global context at multilevel image scales, resulting in richer features with nearly linear computational growth with increasing image sizes. In practice, Swin offers high performance on large-scale image classification and transfer learning to detection and segmentation, making it a popular general-purpose backbone for a large variety of vision tasks while maintaining a low computational complexity [16].

3) ViT:

Google's ViT-Base-Patch16-224-In21k is a Vision Transformer variant with a small footprint that is known for good image-classification performance. It divides the input 224 x 224 image into 16 x 16 patches, linearly projects each patch, maps the positional data, and then feeds the tokens (plus a learnable class token) through stacked transformer encoder layers with multi-head self-attention. Pre-training on ImageNet-21k exposes the model to a wide range of visual diversity, producing representations that can be effectively fine-tuned for downstream tasks. In practice, the product of ViT-Base/16 is a balance between accuracy and efficiency, which can be used as a building block for a variety of recognition issues, such as medical-imaging applications after proper fine-tuning [17].

IV. Experimental Results And Discussions

In Table 2, we provide a head-to-head comparison of classification of transformer baselines and their ensembles. The best performance is obtained from the ViT + DeiT pair with 97.71% accuracy, 97.70% precision, 97.57% recall, and 97.59% F1-score, which demonstrates clear synergy between the two models. Among single models, ViT comes out at the top with 97.25% accuracy, closely followed by DeiT at 95.96% accuracy across nearly equally balanced precision, recall and F1 perks. Swin achieves an accuracy of 93.36% and CaiT an accuracy of 91.08%, showing that this dataset has a preference for global self-attention over strictly windowed or very deep class-attention architectures.

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Table 2. ML and DL models classification results.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
ViT	97.25	97.16	97.09	97.12
DeiT	95.96	95.99	95.92	95.72
Swin	93.36	93.52	93.6	93.07
CaiT	91.08	91.72	91.5	95.71
(ViT+DeiT+ Swin+CaiT)	95.88	95.96	95.59	95.66
ViT + DeiT	97.71	97.7	97.57	97.59
ViT + Swin	96.95	96.94	96.75	96.81
ViT + CaiT	96.8	96.82	96.59	96.65
DeiT + Swin	95.42	95.51	95.13	95.17
DeiT + CaiT	95.42	95.55	95.14	95.34
Swin + CaiT	93.29	93.42	92.85	92.93

A naive all-models ensemble achieves an accuracy of 95.88% and F1-score of 95.66% and underperforms the best pair, indicating that complementarity rather than ensemble size is the key to improvement. Other two-model combinations have a mixed result: ViT + Swin = 96.95%, ViT + CaiT = 96.80%, DeiT + Swin = 95.42%, DeiT + CaiT = 95.42% and Swin + CaiT = 93.29%. All in all, the best balance of accuracy and robustness was achieved by ViT + DeiT while standalone ViT is a good fallback. The most significant observation is that ViT + DeiT ensemble has overall performance of 97.71% which is better than all other pairwise ensembles and even the entire naive ensemble of all four models. This is one of the key points which can be made: performance increase is not caused by the size of an ensemble, but by the complementarity of the models. Although ViT and DeiT have the same transformer structure, they are different in terms of dynamic trainings and attention behaviour, ViT is more focused on the global structure, and DeiT is more focused on fine details since it is trained with the help of distillation. The mixture of these traits results in fewer cases of misclassifications, particularly in tumor types that have a faint structural variation, e.g., glioma and meningioma. The two models used in this study (ViT and DeiT) both have confusion matrices shown in Figure 3(a-d), where the main diagonal represents correct classifications and the off-diagonals represent errors, which shows the confusion between true labels and predicted labels. Both matrices are highly diagonal: the ViT correctly predicted 286 glioma, 292 meningioma, 402 no-tumor and 295 pituitary cases, with its main misclassifications between glioma and meningioma (13 and 7 cases, respectively).

DeiT follows the same trend but sees more glioma being misclassified as meningioma (31), but is unchanged as near-perfect for no-tumor (402) and pituitary (295). Overall, both models work well, with ViT discriminating slightly better between glioma and meningioma than DeiT.

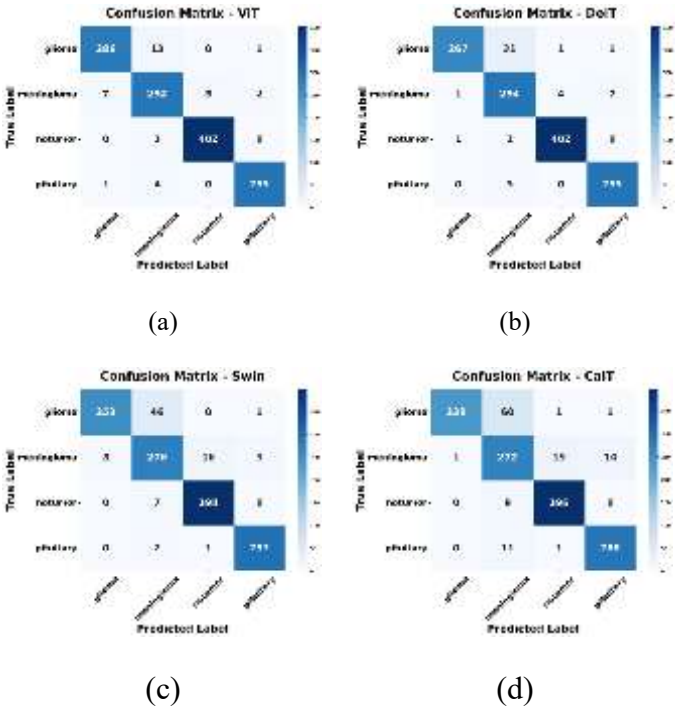


Fig. 3. Confusion matrix of all model.

Figure 4 provides the learning curves of ViT and DeiT. Both models learn quickly during the first few epochs: accuracy increases to approx. 90% and loss decreases significantly, and the curves are starting to flatten after epochs 10-12, which means that we have reached convergence. ViT still keeps getting training accuracy approaching to approx 97%, and the validation accuracy is around 95-96%; the validation loss is a bit rattling around starting from epoch approx 15, indicating slightly overfitting and a minimal generalization gap. DeiT convergence curves are much closer, both for training and validation, and end at approximately 95-96% accuracy with losses steadying down to ~0.10, indicating better generalization. In summary, the majority of the gains are before the 12 epochs, while ViT has the higher peak, and DeiT is more stable across the epochs.

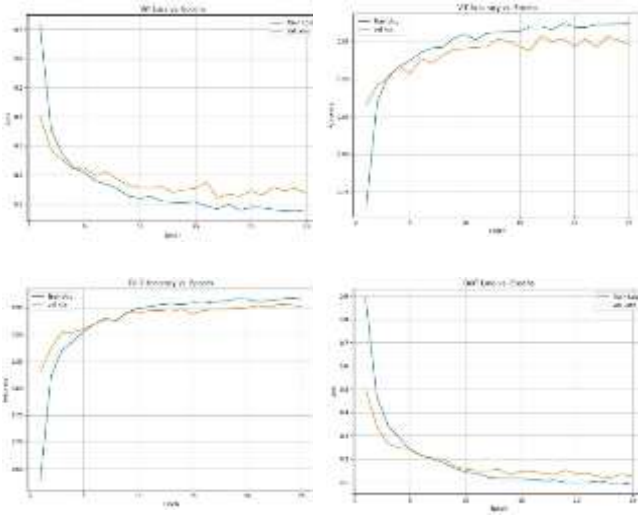


Fig. 4. Training progress of ViT and DeiT (acc/loss).

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Table 3 compares our method to representative MRI brain tumor classification methods. According to Asif et al. [8], Xception was found to have 91.94% accuracy on 3000 images dataset. Ghosal et al. [12] obtained 93.83% using SE-ResNet-101 on 3064 images. Rizwan et al. [11] proposed a GCNN which achieved an accuracy of 97.14% on 3064/516 image datasets. On a larger corpus of 7,023 MRIs, our ViT-DeiT ensemble reaches 97.71% accuracy, which is higher than all accuracies reported in Table 3 surpassing all previous strongest baselines (GCNN) and significantly outperforming SE-ResNet-101 and Xception. The improvement is achieved on a dataset more than twice as large as the one used by Xception and SE-ResNet-101, proving that the improvement is consistent under a larger test set. Indeed, the structure of error appears along the diagonal for our confusion matrices; the lower number of confusions between gliomas and meningiomas and the almost perfect discrimination of no-tumor and pituitary examples, suggest a more reliable classifier. Together, these observations demonstrate the value of ensembling complementary transformer backbones (ViT and DeiT) to provide improved accuracy in brain-tumor MRI classification.

Table 3. Comparing previous work with ours.

References	Dataset	Classifier	Accuracy
[8]	3000	Xception	91.94%
[11]	3064, 516	GCNN	97.14%
[12]	3064	SE-ResNet-101	93.83%
Our Work	7023	ViT + DeiT	97.71%

V. Conclusion And Future Work

A transformer-based model for classification of glioma, meningioma, pituitary and no-tumor from MRI with high classification accuracy and fast inference is presented. First, we construct a four-class MRI corpus with explicit train/validation/test split (7,023 images). Second, nuisance variation is minimized by a modest but consistent image preparation pipeline (resize to 224x224, three channel replication, and normalization with standard augmentations during training). Third, we use a selective fine tuning approach: ImageNet-pretrained ViT and DeiT are frozen to some extent, while the last blocks and a new classification head are trained using AdamW, a cosine learning-rate schedule, and mixed precision. Fourth, we combine predictions using soft-voting and obtain the final decision. On the held-out test set, the ViT+DeiT ensemble obtains 97.71% accuracy (precision 97.70%, recall 97.57%, F1 97.59%), which is better than all other reported accuracies in Table 3. Confusion matrices show well-diagonal with only small glioblastoma-meningioblastoma overlap, and learning curves are steep at the beginning and then reach stable convergence, supporting a consistent and reliable classifier in MRI based brain tumor triage.

Future work will further extend multi-center data, which will increase robustness and generalizability across different patients and scanners. We will also develop the real-time integrated ViT-DeiT ensemble and test it in clinical workflows. Finally, we will extend the framework to other imaging modalities and to other tasks, in order to ensure its versatility beyond the brain-tumor classification.

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